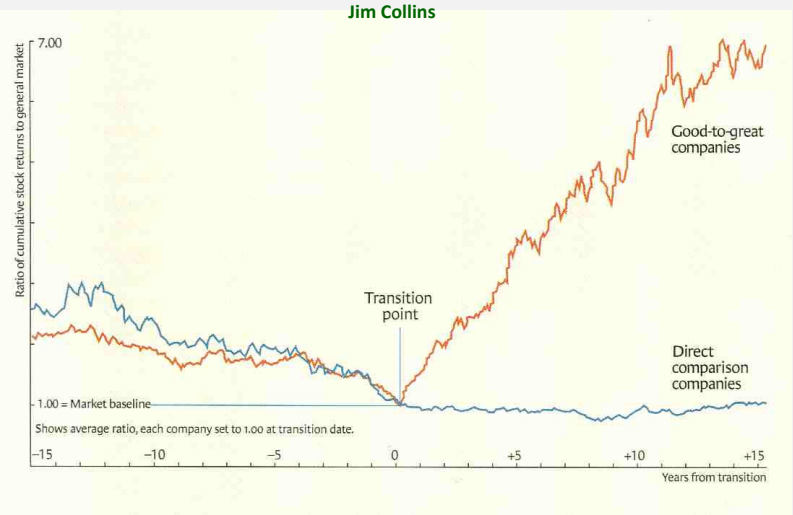
EAS5450 Class #24

"Good to Great"

Jim Collins



1

EAS5450-401 Final Exam

Tuesday, May 6th: 12-2pm
Fisher-Bennett Hall 419

2

EAS5450 Final Exam review sessions

- **Non-Financial Course Content**
  - Thursday, May 1st, 9-11AM
  - Location: Heilmeier Hall – Towne 100
  - Prof. Jeffrey Babin (jbabin@upenn.edu)
  - Submit questions by 12pm, Wednesday, Apr 30th
- **Financial Course Content**
  - Friday, May 2nd, 9-11AM
  - Location: Heilmeier Hall - Towne 100
  - Prof. Tom Cassel
  - No need to submit questions in advance

3

Preparing

- Compile two 2-sided 8½x11 sheets (4 pages) of personal notes: hand-written, cut-and-pasted, or printed. Hard copy, not electronic.
- Review key takeaways from:
  - Assignment study questions
    - Case studies
    - Assigned readings
  - Venture Capital Method & Problem Set
  - Lecture slides and your personal class notes
  - Guest speaker advice and lessons-learned

4

Respondus LockDown Browser

- Required for Final Exam
- Download and configure on your laptop
- Test Quiz in Canvas Quizzes

Be sure to have at least a 2-hour charge on your laptop for the Final Exam. Electric outlets may not be available.

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Name _____

EAS5450 FINAL EXAM

419 Fisher-Bennett Hall
Wednesday, Dec 20th
3:00 – 5:00 pm

The exam has 2 sections:

Section 1 (80 Points). Of the 20 questions in this section, answer any 16 of them. Concise 1-4 sentence responses are sufficient. Where applicable, cite relevant examples from EAS5450 lectures, case studies, articles, and/or guest speakers in your answers. If you answer more than 16 questions, go back and cross out your answers to any questions that you do not want to have included in your exam grade. Graders will grade only the first 16 answers, not the best 16.

Section 2 (20 Points). Answer all 8 of the 8 questions in this section. These involve both a quantitative and qualitative assessment of competing term sheets from two venture capital firms. For the quantitative analysis, you must use the provided calculator -- personal calculators are not allowed. All work must be shown. Graders must be able to see how your quantitative answers are derived. Scratch paper will be provided.

You may refer to 4 pages of personal notes during the exam, otherwise it is closed book. We've stressed the importance of trust since day one of the course. Today you are being trusted to adhere to the *University's Code of Academic Integrity*.

The exam questions have been carefully reviewed to eliminate any ambiguities. Please refrain from asking questions during the exam for clarifications or other guidance -- trust your judgement.

Except for a few students with SDS academic accommodations, the exam will begin at 3 pm on Wednesday, Dec 20th, and will end 2 hours later at 5 pm. If you accidentally hit "submit" before the end and wish to go back to the exam, multiple attempts are allowed within the 2-hour window. Respondus LockDown Browser is required for this Exam.

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Sample Question – Section 1

What are the advantages of raising funds via an IPO and why might an IPO be preferred over an additional round of venture capital?

8

Sample Question – Section 1

What challenges must we confront when scaling a startup?

9

Question Outline – Section 2

- Provided information
  - Company founding and initial capital, shares
  - Business plan fund-raising objectives (timing and amount of investment rounds)
  - Two different investors' term sheets and objectives (timing, investment returns, types of securities, control, etc.)
  - Liquidity event and valuation information
- Question
  - Calculate investors' % acquired, shares, price per share, pre- and post-money valuations, etc.
  - Calculate founders' returns
  - Based on both qualitative and quantitative information, recommend accepting one (or neither) of the two term sheets

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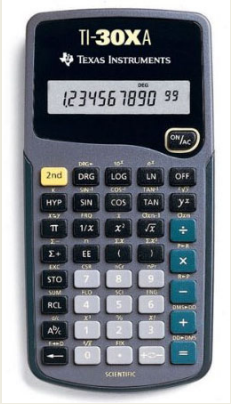
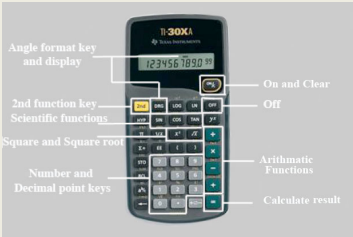
A calculator will be provided for use during the exam (model TI-30XA). Personal calculators may not be used. You may try out a calculator in advance to become familiar with it (recommended).





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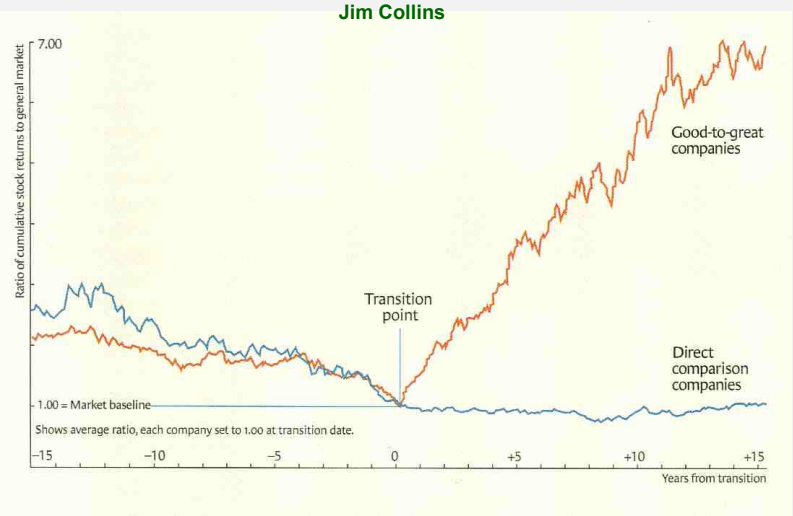
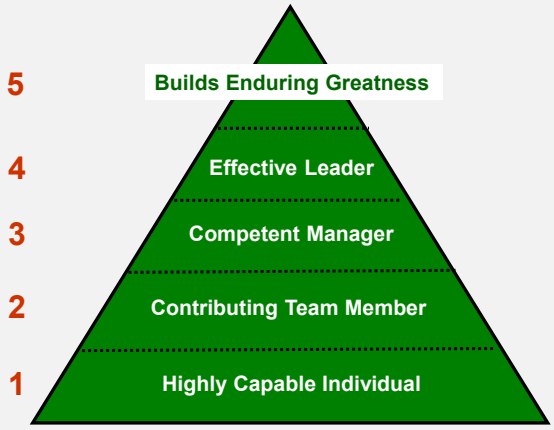
"Good to Great"

Jim Collins



Level 5 Leadership



Collins, Jim. Good to Great. Harper, New York. 2001

"You can accomplish anything in life, provided that you do not mind who gets the credit."

- President Harry Truman

- **Humility**
  - Modest, understated
  - No airs of self-importance
  - No celebrity desire
- **Gracious**
  - Reserved, mild-mannered
  - High emotional IQ
- **Diligent Work Ethic**
  - Workman-like personal style
- **Intense Professional Will**
  - Driven to make the best of anything they touch
  - Fierce, stoic resolve to do what needs to be done
- **Intrinsic Motivator**
  - Rely on inspiring standards...
  - not inspiring charisma.
- **Disciplined**
  - Disciplined thought (confronts brutal facts)
  - Disciplined action
  - Disciplined culture
- **Develop Though Mentors**
  - Self-reflection, personal development
  - Mentors, great teachers, parents
- **Window-Mirror Analogy**
  - Successes
  - Failures

14

Boards (VCs) often feel compelled to recruit outside CEOs, but…

- 10/11 Level-5 leaders came from within the company.
- Non-Great firms recruited outside CEOs @ 6X frequency of Great firms.
- Recruiting high-profile CEOs negatively correlates with company success.

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THE WALL STREET JOURNAL.
WSJ.com

MANAGEMENT | May 29, 2012, 8:55 p.m. ET

Chief Executives Hired Internally Outlast, Outperform Their Rivals

By LESLIE KWOH

As companies renew their focus on internal candidates for rank-and-file positions, boards of directors also may want to look within when hiring a new chief executive.

Chief executives hired from outside a company are twice as likely to be forced out as those promoted from within.

CEOs during that same period, according to consultants Booz & Co.'s 12th annual CEO Succession Study, released on Thursday.

The study analyzed the 2,500 largest publicly traded companies around the world.

Though average tenures for both categories have been declining, insider CEOs tend to hold on to their jobs for five years before they're booted, nearly a year longer than outsider CEOs.

Experts say outsiders are probably more likely to be ousted because they're less familiar with a company's inner workings. That could also explain why companies generally prefer appointing insiders to the corner office in the first place: Four of five new CEOs are hired from inside, the study found.

"It takes a long time to really get to understand how a company works," says Per-Ola Karlsson, Booz's managing director of Europe. "Insiders have a better ability to understand how to affect change within an organization than outsiders typically do."

Another benefit of tapping insiders: They typically deliver better returns for shareholders. The study analyzed the market performance of companies with a chief executive departure between 2009 and 2011 and calculated the company's returns during that CEO's tenure.

Companies with an insider boss outperformed the rest of their respective stock market index by 4.4%, while those with an outsider outperformed their index by just 0.5%.

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THE WALL STREET JOURNAL.
WSJ.com

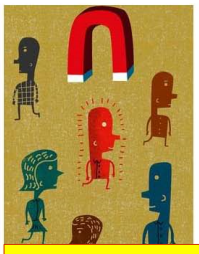
MANAGEMENT | Updated May 29, 2012, 8:26 p.m. ET

An Inside Job: More Firms Opt to Recruit From Within

By RACHEL EMMA SILVERMAN and LAUREN WEBER

What costs more but often works worse? | ***Outside hires.***

Fueled by a conviction that there's plenty of talent in their ranks and backed by research showing that hiring outsiders can lead to costly missteps, firms are ramping up internal hiring efforts and investing in new career sites to boost intra-office movement. So far, those efforts are helping firms cut recruiting costs and retain high performers, companies say.



Cisco Systems Inc. has developed an internal career program, called Talent Connection, that seeks to identify "passive candidates," qualified employees who aren't necessarily looking for a job. The system is designed to help recruiters approach internal candidates the way they do external ones, says Mark Hamberlin, a Cisco vice-president of global staffing.

Since 2010, about half of Cisco's 65,000 employees have created profiles on the website and even more have used it to search for jobs, says Mr. Hamberlin. Talent Connection has saved the company "several million dollars" in search-firm fees and other recruiting costs, while employee surveys show workers' satisfaction with career development has risen by nearly 20 percentage points, he said.

Promoting from within—from the chief executive on down—can deliver more benefits for companies than hiring outside talent, a growing body of research suggests.

One recent study from the University of Pennsylvania's Wharton School found that external hires were paid some 18% more than internal employees in equivalent roles, but fared worse in performance reviews during their first two years on the job.

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Founder-Led Firms

	Average IPO Valuation	3 Years Post-IPO	
		Market-Adjusted Return	Survival Rate
Firms led by founders	$67MM	12% p.a.	73%
Firms led by hired CEOs	$72MM	-26% p.a.	60%

Harvard Business Review, Oct 2002

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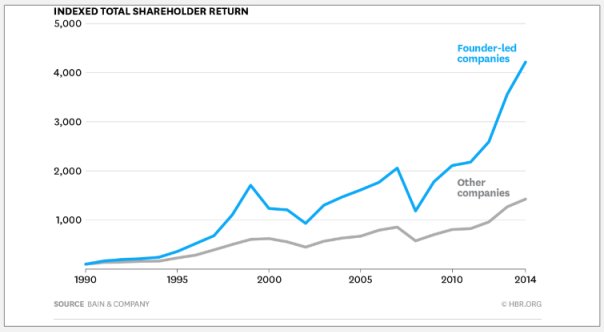
Founder-Led Companies Outperform the Rest – Why?



3 HARD-EDGED PRACTICES & UNDERLYING ATTITUDES…

- **Insurgency** - Disrupt industry norms on behalf of underserved customers.
- **Heroes for Front Line** – Obsession with details and the entrepreneurial culture.
- **Owner's Mindset** – Act with speed and take responsibility for risk and cost.

Zook, C., Bain & Co., HBR March 2016

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Founder-Led Companies Lead the Way

20

"We've often failed when importing CEOs … better to let the entrepreneur run the show than to destroy the culture."

Doug Alexander, Managing Director, Internet Capital Group, Mar 2, 2006

Andy Rachleff, Managing Partner, Benchmark Capital, Nov 9, 2004

"Engineers need to be CEOs of technology companies. Quality VCs don't want to control and run companies."

"Don't let others define your limitations."

Kathy Crothall, Founder & CEO, Animas Corp, Mar 3, 2011

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MIPS Computer Systems

Compact, fast microprocessor
- 10X faster
- Whole product:
  - RISC architecture +
  - UNIX operating system

Team
- John Hennessy, Founder
- Stanford faculty & researchers
- HP, IBM, INTEL defectors
- Vaemond Crane, CEO

Cash
- DARPA funding
- $20M Series A: Mayfield VC
  - 2 board seats
  - Executive recruitment

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Was MIPS' $8 million revenue goal an effective mission for the company?

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Wall Street Journal
March 29, 2001 - Page 1

Phone Numbers

Behind Lucent's Woes: All-Out Revenue Goal And Pressure to Meet It

CEO, Sales Official Clashed Before Spate of Revisions That Tarnished Credibility

Credits and Channel Stuffing

By DENNIS K. BERMAN
And REBECCA BLUMENSTEIN
Staff Reporters of THE WALL STREET JOURNAL

The pressure was building day by day inside Lucent Technologies Inc. last summer. Chief Executive Richard McGinn was under the gun and that meant the sales staff was too.

Mr. McGinn had turned the former equipment-making arm of AT&T Corp. into a Wall Street star by increasing its sales at a double-digit pace. But in 2000 he had twice disappointed investors, and he was determined not to let it happen again. Monday-morning conference calls among the sales troops could get downright nasty.

During one on Aug. 14, North American sales chief Nina Aversano warned Mr. McGinn that big customers were pulling back, say people familiar with the call. Sensing an economic slowdown, she said it appeared her team would miss Lucent's just-reduced quarterly revenue expectation. The CEO "went ballistic," wrote a member of her team, Marc Schweig, in an e-mail later. "It was very painful for Nina and all of us." Mr. Schweig told fellow sales staffers that "we need to redouble our efforts at any possible deals . . . no holds barred."

What followed was a frenzy of deal-making as Lucent salespeople went all-out to meet the revenue targets. They almost did, falling just $100 million short of a nearly $6 billion revenue goal when the fiscal fourth quarter ended Sept. 30.

But in the process, Lucent borrowed from the future. It promised customers a host of discounts, one-time credits and other incentives certain to eat into future sales. It booked sales for goods that were shipped merely to distribution channels and not to real customers—"channel stuffing," as it's called. Besides making the fourth-quarter figures questionable, Ms. Aversano later calculated, these gimmicks, along with customers' growing caution and Lucent's reduced financing of sales, meant that North American revenue in the fiscal year beginning Oct. 1 would miss projections by nearly 20%.

On Oct. 9, Ms. Aversano went to see Mr. McGinn and told him so.

It set off an uproar. The next day, Lucent announced that fourth-quarter results would be below goals and would "impact and lower its guidance for 2001." The stock plunged. Ms. Aversano resigned. Two weeks later, Mr. McGinn was fired.

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Lessons from Spoiled Startup…

1. Pretend you don't have it.
2. Stay strategic.
3. Find customers first.
4. Be sure you have a product.

25

Robert Miller

Level 5 Leadership

Never did we lose faith in the basic technology. Bob was the only one we interviewed who shared our vision for the future: design and produce a RISC architecture capable of running the UNIX operating system. In addition, produce a new generation of system software and compilers which will make our machine's RISC design transparent to the applications programmer. Bob also had the required interpersonal skills and operational competence.

A. Grant Heidrich, General Partner
Mayfield Venture Capital

- Humble, driven, focused
- Driven by belief in the technology
- Focused on market-driven products
- Restored an effective vision & culture

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MIPS Positioning Strategy under Miller…

Positioning = Segmentation + Differentiation

1. Define ideal customer who wants to buy it.
2. Develop a strategy to reach them.
3. Convince the customer you're better.

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What should Robert Miller do?

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MIPS Computer Systems
Income Statement

	(Disguised Figures)	
	1986 Year	1987 1st Qtr
Revenue		
Hardware	$5,065	$2,176
Software	$2,036	$473
Net Sales	$7,101	$2,649
Cost of Sales		
Manufacturing	$1,250	$605
Overhead Absorption	($394)	($140)
Product cost	$2,291	$1,024
Other Cost	$386	$465
Cost of Sales	$3,533	$1,955
Gross Margin	$3,568	$694
Operating Expenses		
Engineering	$6,773	$1,904
Marketing/Sales	$1,724	$808
Administration	$3,426	$1,522
Total Operating Expenses	$11,923	$4,234
Operating Profit	($8,356)	($3,540)
Other (Income)/Expenses	($401)	$68
Profit Before Taxes	($7,955)	($3,607)
Net Profit After Taxes	($7,955)	($3,607)

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MIPS Computer Systems

Balance Sheet

(Disguised Figures)

	Dec. 31, 1986	Apr. 30, 1987
Current assets:		
Cash and Short term investments	$2,988	$777
Accounts Receivable, net	$5,210	$3,326
Inventories	$1,649	$2,422
Other current assets	$228	$205
Total current assets	**$10,075**	**$6,730**
Property Plant and Equipment		
Machinery and Equipment	$6,300	$6,600
Furniture and fixtures	$1,806	
Other Assets		$1,121
Leasehold improvements	$121	
Accumulated Depreciation	$1,110	$1,805
Net property and equipment	**$7,117**	**$5,916**
Restircted cash	$1,100	$1,105
Deposites and other assets	$221	
Total Assets	**$18,513**	**$13,751**
Liabilities and Shareholders Equity		
Current liabilities:		
Accounts Payable	$6,009	$3,451
Current portion of LTD	$181	$1,609
Other Current Liabilities	$2,328	
Total Current Liabilities	**$8,518**	**$5,060**
Deferred revenue	$984	$1,285
Long term debt	$363	$2,282
Other Long Term Liabilities	$1,844	
Total Liabilities	**$11,709**	**$8,627**
Shareholders equity		
Preferred Stock	$20,336	$20,336
Common stock	$377	$666
Subordinated Debt		$1,800
Accumulated Deficit	($13,909)	($17,678)
Total Shareholders Equity	**$6,804**	**$5,125**
Liabilities and Shareholders Equity	**$18,513**	**$13,752**

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Mission

To make the MIPS microprocessor the most pervasive RISC microprocessor in the entire world.

MIPS Everywhere®

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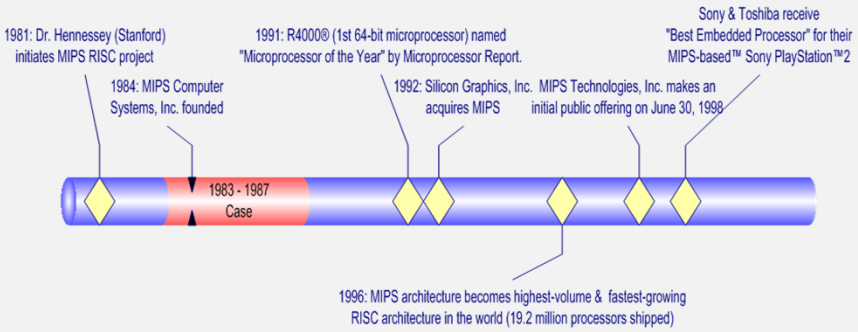
MIPS Timeline



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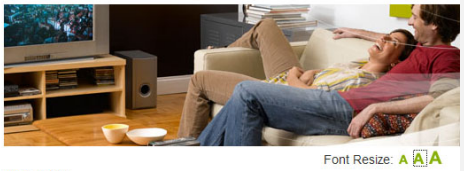
MIPS TECHNOLOGIES



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About Us

Technology Innovation to IP Leadership

MIPS Technologies is a leading provider of industry-standard processor architectures and cores that power some of the world's most popular products for the home entertainment, communications, networking and portable multimedia markets.

For more than two decades, MIPS Technologies has been a leader and innovator in the worldwide embedded semiconductor market. At the heart of MIPS is its architecture, developed 20 years ago by Stanford University engineering Professor John Hennessy-now president of Stanford University. Hennessy took the lead in RISC processing and created an elegant, streamlined architecture with a scalability that has met the demands of generations of applications, preserving the wealth of development tools and software that support them. Today, the MIPS® architecture is an industry standard and the performance leader within the embedded industry.

There are millions of MIPS-Based™ products around the world, including broadband devices from Linksys, digital cameras, DTVs and entertainment systems from Sony, DVD recordable devices from Pioneer, digital set-top boxes from Motorola, network routers from Cisco, 32-bit microcontrollers from Microchip Technology and laser printers from Hewlett-Packard.

www.mips.com (2012)

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MIPS Market Position

Segment	Market Share
Cable Modem	95%
Internet Backbone	40%
DSL Modem	52%
VoIP	72%
WLAN Access	55%
DVD Recorder	70%
Cable STB	76%
Satellite STB	30%
Digital TV	59%
Color Laser Printer	62%
Commercial Color Copiers	73%

http://www.mips.com/content/Corporate/AboutUs (2007)

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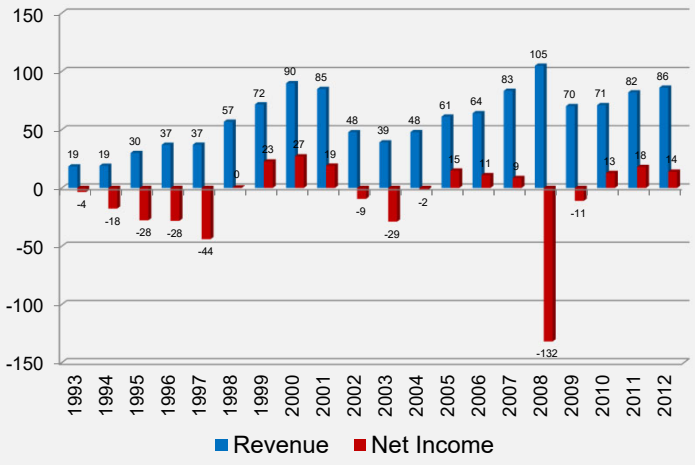
MIPS Performance

($Millions – FY Ending June 29th)



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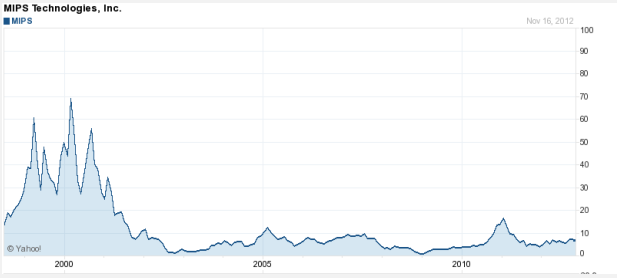
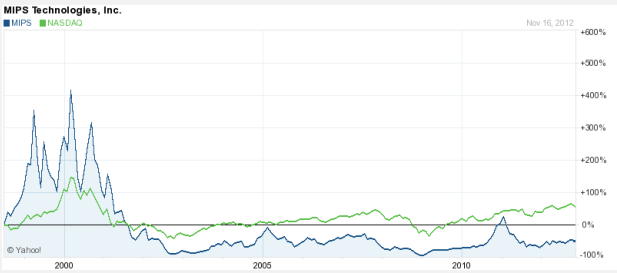
MIPS Stock Performance





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Tom's Hardware > News > Components > CPU > MIPS Acquired by Imagination Technologies, Most Pa

MIPS Acquired by Imagination Technologies, Most Patents Go To Industry Organization

9:00 AM - November 7, 2012 by Wolfgang Gruener - source: MIPS

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We are not exactly surprised that MIPS has been acquired since the company was shopping for a buyer for several months.

Imagination agreed to acquire the company for about $60 million in cash. The assets include the operating business as well as 82 of a total of 580 patents, and a perpetual license to the remaining 498 patents. At the end of October, MIPS reported cash and investment reserves of about $130 million, which will be retained to cover "tax and other liabilities". The leftovers will be distributed to the MIPS shareholders.

Zoom

498 patents were acquired by Bridge Crossing, a new entity created by Allied Security Trust (AST) that apparently has the purpose to manage the patent portfolio. AST's member companies are Avaya, HP, IBM, Intel, Motorola, Oracle, Philips, and Research in Motion, but there was no information which of those companies participate in Bridge Crossing. What we know is that ARM has identified itself as a leading member of Bridge Crossing and said that it paid $167.5 million of a total of $350 million for the patent portfolio.

"ARM is a leading participant in this consortium which presents an opportunity for companies to neutralize any potential infringement risk from these patents in the further development of advanced embedded technology," said ARM CEO Warren East in a prepared statement. "Litigation is expensive and time-consuming and, in this case, a collective approach with other major industry players was the best way to remove that risk."

MIPS CEO Sandeep Vij said that, "after a thorough review of a wide range of alternatives to enhance shareholder value, our board of directors concluded that the best alternative for the Company would be the sale of patent properties to a consortium formed by AST, and a sale of the Company to Imagination."

MIPS' history dates back to 1981, when Stanford's John Hennessy began working on the MIPS RISC architecture project. MIPS was founded in 1984.

http://www.tomshardware.com/news/mips-acquisition-arm-patent,18950.html

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